

Test 1

Speaking Math — Sets — Functions

Name: _____

Time: 1.5 hours

Instructions

Use a **pencil**. **Show all your work and explain your reasoning** — you are a mathematician, so a bare answer with no justification will not earn points. Partial credit will be given, so explain your thought process as much as possible, even if you can't answer it. When a question asks you to *explain in words*, write in full sentences. The test is worth 100 points (plus a 6-point bonus). You have 1.5 hours; manage your time, and do not get stuck on any one problem. If something is unclear, ask Leo.

Part 1 — Speaking Math

1. [6 pts] Classify each of the following as an **expression**, an **equation**, or a **formula**.

(a) $2x - 7$ _____

(b) $y = 5x$ (assume no context is given) _____

(c) Volume = Base $\cdot$ Height _____

(d) $3x + 1 = 10$ _____

(e) 12 _____

In one sentence: what makes something a **formula** rather than an equation?

2. [10 pts] Consider the expression $5x^3 - x^2y + 8x - 4$.

(a) How many terms does it have?

Answer: _____

(b) Fill in the table.

Term	Coefficient	Variable(s)	Exponent(s)
$5x^3$			
$-x^2y$			
$8x$			
-4			

(c) Mark each True or False. If false, write the corrected version.

T / F The coefficient of the x -term is 1.

T / F -4 is a term.

T / F $5x^3$ and $8x$ are like terms.

3. [6 pts] Speaking math.

(a) Read this aloud and write down the complete English sentence you would say:

$$(x + 2)(x - 3)^2 + 7$$

(b) Write the algebraic expression for: “the quantity three x minus one, over the quantity x plus two.”

Part 2 — Sets

Recall

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \quad (\text{the integers})$$

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\} \quad (\text{the natural numbers})$$

$$\mathbb{R} = \{\dots, -3.24, 0, 1.99, \sqrt{2}, \pi, 10, \dots\} \quad (\text{the real numbers})$$

4. [8 pts] Let $S = \{3, 6, 9, 12\}$.

(a) Fill in $\in$ or $\notin$: 6 ____ S 7 ____ S 12 ____ S 0 ____ S

(b) Write S in set-builder notation (you can use words for the condition).

(c) Write each in roster notation.

(i) $\{x \in \mathbb{Z} \mid -2 \leq x \leq 2\}$ = _____

(ii) $\{x \in \mathbb{N} \mid x^2 < 5\}$ = _____

5. [4 pts] Subsets.

(a) List **all** subsets of $\{p, q\}$ Which one is the **improper** subset?

6. [8 pts] Let $U = \{1, 2, 3, \dots, 10\}$, $A = \{2, 3, 5, 7\}$, $B = \{1, 2, 3, 4\}$. Fill out the following chart:

Expression	Result
$A \cup B$	
$A \cap B$	
$A \setminus B$	
A^c (in U)	
$ A $	
$ A \cup B $	

7. [4 pts] Real-world sets.

(a) In a class of 28 students, 15 play guitar, 12 play piano, and 5 play **both**. How many play **at least one** instrument? Show your work.

Part 3 — Functions

8. [6 pts] Let $f(x) = 3x - 5$.

(a) Fill in the table.

x	0	2	-1
$f(x)$			

(b) Find x such that $f(x) = 7$.

Answer: _____

(c) Write f as a set of **ordered pairs** for the inputs $\{0, 2, -1\}$.

9. [8 pts] For each rule, decide whether it is a valid function from A to B . If it is, state whether it is **injective** and whether it is **surjective**. If it is not a function, explain why not.

(a) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$: $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 4$.

(b) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$: $1 \mapsto 2$, $2 \mapsto 3$, $1 \mapsto 4$, $3 \mapsto 2$.

(c) $A = \{1, 2, 3\}$, $B = \{a, b\}$: $1 \mapsto a$, $2 \mapsto b$, $3 \mapsto a$.

(d) $A = \{1, 2, 3\}$, $B = \{a, b, c\}$: $1 \mapsto a$, $2 \mapsto c$.

10. [4 pts] Domain, codomain, range.

For $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 + 1$, state the domain, the codomain, and the range.

11. [8 pts] Injective, surjective, bijective.

(a) Write a function $\{1, 2, 3\} \rightarrow \{a, b, c\}$ that is **bijective**.

(b) Write a function $\mathbb{Z} \rightarrow \mathbb{Z}$ that is **injective but not surjective**.

12. [8 pts] Prove that $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 4x + 1$ is bijective.

- 13.** [6 pts] Prove that $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2 + 1$ is **not** bijective.

14. [6 pts] Real-world application. A water park charges \$8 to get in, plus \$4 for each ride ticket you buy. You buy r ride tickets.

(a) Write a function $C(r)$ for the total cost (in dollars) of your visit. Give its signature ($C: A \rightarrow B$) and its rule.

(b) In one sentence, what would it mean **in this context** for C to be injective?

(c) Is C surjective onto all positive integers?

15. [8 pts] Big ideas about infinity. Answer each in words.

(a) We cannot finish counting an infinite set. So how do we decide whether two infinite sets have the **same size**?

(b) Who is the mathematician famous for the **diagonal argument** showing that $\mathbb{R}$ is larger than $\mathbb{N}$?

(c) What makes the **Collatz Conjecture** a conjecture, but not a theorem?

BONUS: [+6 pts] Determine if $\mathbb{N}$ and $\mathbb{Z}$ are the same size. Explain your answer.